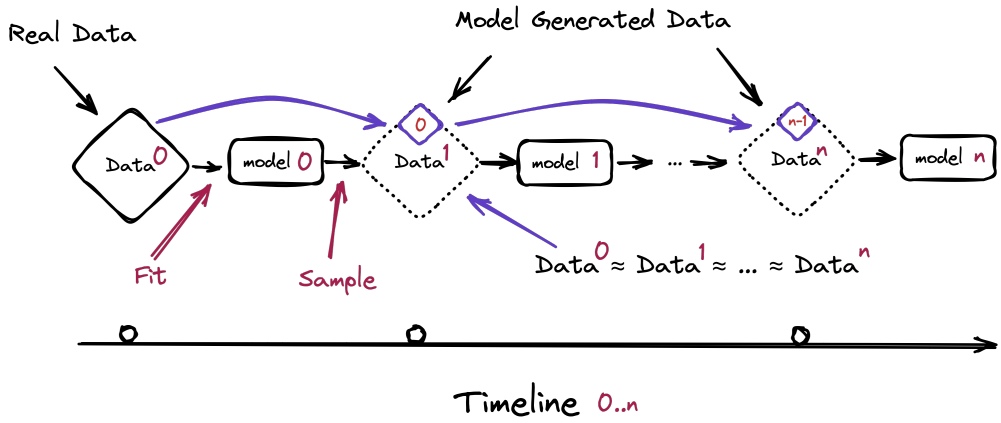

Figure 2: The high-level description of the feedback mechanism in the learning process. Here, data are assumed to be human-curated and start off clean; then model 0 is trained and data are sampled from it; at step n, data are added to the overall data from step $n-1$, and this ensemble is used to train model n. Data obtained with Monte Carlo sampling should ideally be statistically close to the original, provided *fitting* and *sampling* procedures are perfect. This process depicts what happens in real life with the Internet – model-generated data become pervasive.

3.1 Causes of *model collapse*

There are two main causes for *model collapse*, one primary and one secondary, which we describe now. Further mathematical intuition is provided in Section 4 to explain how these give rise to the errors observed, how different sources can compound and how we can quantify the average model divergence rate.

- *Statistical approximation error* – this is the primary type of error, which arises due to the number of samples being finite, and disappears as the number of samples tends to infinity. This occurs due to a non-zero probability that information can get lost at every step of re-sampling. Figure 12 shows an example of an approximation error. Here, a single-dimensional Gaussian is being approximated from a finite number of samples. Despite using a very large number of points, the errors remain significant; with 10^7 samples we estimate the mean to be $0.00024899 \pm 1.89382984e^{-4}$, when the true value is 0.

- *Functional approximation error* – this is a secondary type of error, which stems from our function approximators being insufficiently expressive (or sometimes too expressive outside of the original distribution support [Nguyen et al., 2015]). It is well known that neural networks are universal functional approximators in the limit, but in practice this is not always true. In particular, a neural network can introduce non-zero likelihood outside of the support of the original distribution. A simple example of this error is if we were to try fitting a mixture of two Gaussians with a single Gaussian. Even if we have perfect information about the data distribution, model errors will be inevitable. It is important to also note that in the absence of statistical error, functional approximation error only occurs at the first generation. Once the new distribution belongs to the image of functional approximator, it remains exactly the same over the generations.

Each of the above can cause *model collapse* to get worse or better. Better approximation power can even be a double-edged sword – better expressiveness may counteract statistical noise, resulting in a good approximation of the true distribution, but it can equally compound this noise. More often then not, we get a cascading effect where combined individual inaccuracy causes the overall error to grow. Overfitting the density model will cause the model to extrapolate incorrectly and might give high density to low-density regions not covered in the training set support; these will then be sampled with arbitrary frequency.

It is worth mentioning that modern computers also have a further computational error coming from the way floating point numbers are represented. This error is not evenly spread across different floating point ranges, making it hard to estimate the precise value of a given number. Such errors are smaller in magnitude and are fixable with more precise hardware, making them less influential on *model collapse*.

4 Theoretical intuition

In this section we aim to provide a theoretical intuition for the phenomenon of *model collapse*. We argue that the process of *model collapse* is universal among generative models that recursively train on data generated by previous generations. We construct toy mathematical models, which prove to be simple enough to provide analytical expressions for quantities of interest, but also portray the phenomenon of *model collapse*. We aim to quantify how different sources of error can affect the overall end approximation of the original distribution. As discussed in Section 3.1, there are two main sources we are interested in – *statistical* error and *functional* error. Since in the real world one rarely has infinite samples, quantifying the functional approximation error alone is of little interest for discussion of *model collapse*. Therefore, we will examine two simple cases: a discrete distribution in the absence of functional approximation error and a single dimensional Gaussian case, which portrays how functional approximation error can compound with statistical error.

The overall stochastic process we are going to be considering (which we call *Learning with Generational Data*) is the following. Assume that at generation i we have a dataset $\mathcal{D}_i$ comprising of i.i.d. random variables X_j^i, where $j \in \{1, \dots, M_i\}$ denotes the sample number at generation i and $M_i \geq 2$. We will denote the distribution of X^i as p_i. Here we assume that p_0 denotes the original distribution, from which the data comes from. Going from generation i to generation $i+1$, we aim to estimate the distribution of samples in $\mathcal{D}_i$, with an approximation $p_{\theta_{i+1}}$. This step is what we refer to as functional approximation $\mathcal{F}_\theta : p_i \to p_{\theta_{i+1}}$. We then resample the dataset $\mathcal{D}_{i+1}$ from the distribution $p_{i+1} = \alpha_i p_{\theta_{i+1}} + \beta_i p_i + \gamma_i p_0$, with non-negative parameters $\alpha_i, \beta_i, \gamma_i$ summing up to 1, *i.e.* they represent proportions of data used from different generations. This corresponds to a mixing of data coming from the original distribution (γ_i), data used by the previous generation (β_i) and data generated by the new model (α_i). We refer to this as the sampling step. For the mathematical models to come, we consider $\alpha_i = \gamma_i = 0$ *i.e.* data only from a single step is used, while numerical experiments are performed on more realistic choices of parameters.

4.1 Discrete distributions with exact approximation

In this subsection we consider a discrete probability distribution, which is represented by a histogram, *e.g.* as shown on Figure 3. In what follows we consider the stochastic process in absence of functional approximation error, *i.e.* $\mathcal{F}(p) = p$. In this case, *model collapse* arises only due to statistical errors from the sampling step. At first, the tails (low probability events) begin to disappear due to low probability of sampling them, and over time the distribution becomes a delta function. Denoting the sample size as M, if we consider state i with probability $q \leq \frac{1}{M}$, the expected number of samples with value i coming from those events will be less than 1, which means that in practice we will lose information about them. This is portrayed on Figure 3, where infrequent events get cut off. Considering more generally some state i with probability q, using standard conditional probability one can show that the probability of losing information (*i.e.* sampling no data at some generation) is equal to $1 - q$. But this in turn means that we must converge to a delta function positioned at some state, and the probability of ending up at a certain state is equal to the probability of sampling said state from the original distribution.

But how do we show directly that this process is going to turn our distribution into a delta function? By considering the process as going from $\mathbf{X}^i \to \mathcal{F}_\theta \to p_{i+1} \to \mathbf{X}^{i+1}$, we see that this forms a Markov Chain, as $\mathbf{X}^{i+1}$ only depends on $\mathbf{X}^i$. Furthermore, if all the X_j^i have the same value, then at the next generation the approximated distribution will be exactly a delta function, and therefore all of X_j^{i+1} will also have the same value. This implies that the Markov chain contains at least one absorbing state, and therefore with probability 1 it will converge to one of the absorbing states. This is a well-known fact, of which a proof is provided in Appendix A.1. For this chain, the only absorbing states are those corresponding to delta functions. As a result, as we follow the progress of *model collapse*, we are guaranteed to end up in a constant state, having lost all the information of the original distribution when the chain is absorbed.[3] Based on the discussion above we see how both early and late stage *model collapse* must arise in the case of discrete distributions with perfect functional approximation.

4.2 Single dimensional Gaussian

Following the discussion about discrete distributions, we now move on to considering how both functional approximation error and sampling error can compound (or cancel out) the process of *model collapse*.

To demonstrate this, consider a single dimensional Gaussian $X^0 \sim \mathcal{N}(\mu, \sigma^2)$. If we have full faith in the data we observe, the functional approximation involves estimating sample mean and variance and fitting a single dimensional

[3] This argument also works in general due to floating point representations being discrete, making the Markov Chain over the parameters of the model discrete. Thus as long as the model parameterisation allows for delta functions, we *will* get to it, as due to sampling errors the only possible absorbing states are delta functions.

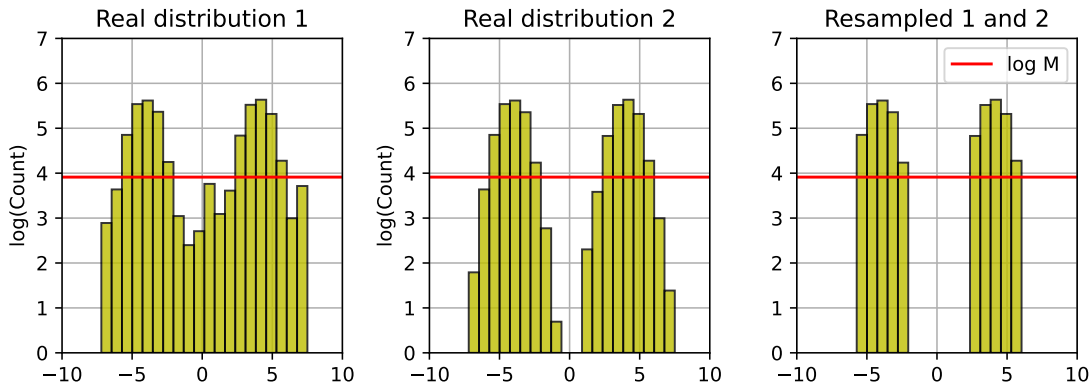

Figure 3: Shown in the middle is a histogram plot of samples from a Gaussian mixture with means $(-4, 4)$ and variances of 1. To the left of it is a similar distribution, but with 'fatter' tails, and on the right the same histograms are shown, but with low probability events being cut off due to finite resampling. Although distributions 1 and 2 are very different, when resampled (only assuming the expected behaviour), the tails get cut off, leading to the same observed distribution. In this case this is all states with probability less than $1/M$, or equivalently, bins with $\log \texttt{Count} \leq \log M$.

Gaussian. We can estimate them using the unbiased sample mean and variance estimators:

$$\mu_{i+1} = \frac{1}{M_i} \sum_j X_j^i; \quad \sigma_{i+1}^2 = \frac{1}{M_i - 1} \sum_j (X_j^i - \mu_{i+1})^2. \tag{1}$$

Note here, that if we were to use maximum likelihood estimation, we would instead arrive at a biased variance estimator. With these estimates, the functional approximation step simply corresponds to considering a normal distribution with these parameters, which we can sample from:

$$X_j^{i+1} | \mu_{i+1},\ \sigma_{i+1} \sim \mathcal{N}(\mu_{i+1}, \sigma_{i+1}^2). \tag{2}$$

This provides us with the conditional distribution of X_j^i, which allows us to calculate the full distribution of X_j^i. From Equation (3), we see that even after the first approximation, the distribution of X_j^i is no longer normal, it follows a variance-gamma distribution [Fischer et al., 2023]. However, instead of writing the probability density function at each generation, we can explicitly construct them in terms of independent random variables. In particular, it is well known [Cochran, 1934] that μ_1 and σ_1 are independent, with $\mu_1 \sim \mathcal{N}(\mu, \frac{\sigma^2}{M_0})$ and $(M_0 - 1)\sigma_1^2 \sim \sigma^2 \Gamma(\frac{M_0 - 1}{2}, \frac{1}{2})$. In what follows we will denote with Z random variables that are distributed with $\mathcal{N}(0, 1)$ and with S^i random variables that are distributed with $\frac{1}{M_{i-1}-1}\Gamma(\frac{M_{i-1}-1}{2}, \frac{1}{2})$.

$$\begin{aligned} X_j^0 &= \mu + \sigma Z_j^0; \quad X_j^1 = \mu + \frac{\sigma}{\sqrt{M_0}} Z^1 + \sigma \sqrt{S^1} Z_j^1; \quad \dots \\ X_j^n &= \mu + \frac{\sigma}{\sqrt{M_0}} Z^1 + \frac{\sigma}{\sqrt{M_1}} \sqrt{S^1} Z^2 + \dots + \frac{\sigma}{\sqrt{M_{n-1}}} \sqrt{S^1 \times \dots \times S^{n-1}} Z^n + \sigma \sqrt{S^1 \times \dots \times S^n} Z_j^n. \end{aligned} \tag{3}$$

These are not joint distributions, as Z^n and S^n depend directly on Z_j^{n-1}, but when considering X_j^n on its own the formula above provides all the information about the full distribution.

The first thing we may try calculating is the variance. It is possible to find its exact value, but the mean and variance of the square root of gamma distribution are expressed in terms of gamma functions, making the result quite clunky. In what follows, we will expand everything to second order in each of $(1/M_i)$ as we assume each sample size to be large (in practice this becomes quite accurate after $M \sim 100$). We then find that

$$\frac{1}{\sigma^2} \operatorname{Var}(X_j^n) = \frac{1}{M_0} + \frac{1}{M_1} + \dots + \frac{1}{M_{n-1}} + 1 + \mathcal{O}(2).$$

And if we were to assume that $M_i = M$ are constant, we would find that:

$$\operatorname{Var}(X_j^n) = \sigma^2 \left(1 + \frac{n}{M}\right); \quad \mathbb{E}(X_j^n) = \mu.$$